

Product Customizer Canvas — Complete Customization Feature List

1. Canvas Core Controls

প্রতিটি customizable product-এর জন্য basic canvas controls:

- Select object
 - Drag / Move object
 - Resize object
 - Rotate object
 - Delete object
 - Duplicate object
 - Copy
 - Paste
 - Cut
 - Lock object
 - Unlock object
 - Hide object
 - Show object
 - Flip horizontal
 - Flip vertical
 - Reset transformation
 - Bring forward
 - Send backward
 - Bring to front
 - Send to back
 - Center horizontally
 - Center vertically
 - Align left
 - Align right
 - Align top
 - Align bottom
 - Multi-select objects
 - Group objects
 - Ungroup objects
-

2. Canvas Navigation

Canvas workspace-এর জন্য:

- Zoom in
 - Zoom out
 - Zoom percentage
 - Fit to screen
 - Actual size / 100%
 - Reset zoom
 - Pan canvas
 - Hand tool
 - Selection tool
 - Full-screen editor
 - Mouse-wheel zoom
 - Touch pinch-to-zoom
 - Touch pan
 - Responsive canvas viewport
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3. Undo / Redo / History

- Undo
 - Redo
 - Keyboard `Ctrl + Z`
 - Keyboard `Ctrl + Shift + Z`
 - History stack
 - Object-change snapshots
 - Prevent duplicate history entries
 - Restore previous design state
 - Reset entire design
 - Version restore
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4. Text Tool

Canvas-এ user text add করতে পারবে।

Basic Text

- Add text
- Add heading
- Add subheading
- Add paragraph
- Edit text directly on canvas
- Multi-line text

Typography

- Font family
- Font search
- Font categories
- Recently used fonts
- Favorite fonts
- Font size
- Font weight
- Bold
- Italic
- Underline
- Strikethrough
- Uppercase
- Lowercase
- Capitalize

Alignment

- Left align
- Center align
- Right align
- Justify

Spacing

- Letter spacing
- Line height
- Paragraph spacing

Text Color

- Text color
- Hex input
- RGB

- Color picker
- Recently used colors
- Brand colors

Text Effects

- Text shadow
- Shadow blur
- Shadow offset
- Shadow color
- Text outline
- Outline width
- Outline color
- Background behind text
- Text opacity

Advanced Text

- Curved text
 - Arc text
 - Circle text
 - Text bend amount
 - Vertical text
 - Text rotation
 - Text warp — advanced
 - Text along path — advanced
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5. Image Upload Tool

Upload

- Upload PNG
- Upload JPG/JPEG
- Upload WEBP
- Upload SVG
- Drag-and-drop upload
- Upload progress
- Multiple upload
- Uploaded image library

Image Controls

- Move
- Resize
- Rotate
- Crop
- Replace image
- Delete
- Duplicate
- Flip horizontal
- Flip vertical
- Opacity

Image Crop

- Free crop
- Square crop
- Circle crop
- Custom aspect ratio
- Product-area crop
- Reset crop

Image Adjustments

- Brightness
- Contrast
- Saturation
- Grayscale
- Sepia
- Blur
- Sharpen
- Exposure — advanced
- Hue — advanced

Image Validation

- Low-resolution warning
- DPI warning
- File-size warning
- Unsupported-format warning
- Print-quality indicator

Example:

Excellent Quality

Good Quality

Low Resolution

Not Recommended

6. Background Removal

Advanced image feature:

- Remove background
- Restore original background
- Background-removal loading state
- Compare before/after
- Replace processed image
- Background-removal pricing support

Backend later worker/API service-এর মাধ্যমে process করতে পারবে।

7. Shapes Tool

Available shapes:

- Rectangle
- Rounded rectangle
- Square
- Circle
- Ellipse
- Triangle
- Polygon
- Star
- Heart
- Arrow
- Line
- Custom SVG shape

Shape controls:

- Fill color
- Border color
- Border width
- Border style

- Corner radius
 - Opacity
 - Shadow
 - Rotation
-

8. Lines & Dividers

- Straight line
 - Dashed line
 - Dotted line
 - Arrow line
 - Double arrow
 - Line thickness
 - Line color
 - Line opacity
 - Line caps
-

9. Clipart / Elements

Admin-managed resource library |

Categories:

- Icons
- Symbols
- Decorative graphics
- Illustrations
- Badges
- Frames
- Borders
- Patterns
- Stickers

Features:

- Search elements
- Categories
- Recently used

- Favorites
 - Admin-uploaded clipart
 - SVG recoloring
 - Element opacity
 - Element rotation
 - Element layering
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10. QR Code Generator

- Add QR code
- URL QR
- Text QR
- Phone QR
- Email QR
- Wi-Fi QR — optional
- QR foreground color
- QR background color
- QR size
- Error correction
- QR validation

Useful for:

Business Cards

Menus

Labels

Tickets

Packaging

Event Materials

11. Barcode Generator

Optional but useful:

- Barcode
- SKU barcode
- Code128
- EAN

- UPC
 - Barcode number
 - Barcode size
 - Barcode color
-

12. Background Tool

Canvas background customization:

- Solid color
 - Gradient
 - Background image
 - Pattern background
 - Transparent background
 - Reset background
 - Lock background
 - Background opacity
-

13. Pattern Tool

- Pattern library
- Repeat pattern
- Pattern scale
- Pattern rotation
- Pattern spacing
- Pattern color
- Upload custom pattern

Useful for:

Apparel
Packaging
Tote Bags
Mugs
Phone Cases
Gift Products

14. Layers Panel

Professional editor-এর mandatory feature।

- Show all layers
 - Layer thumbnail
 - Layer name
 - Rename layer
 - Select layer
 - Reorder via drag
 - Hide/show
 - Lock/unlock
 - Duplicate
 - Delete
 - Group layers
 - Ungroup
 - Bring front
 - Send back
-

15. Smart Alignment & Snapping

- Horizontal center guide
 - Vertical center guide
 - Edge snapping
 - Object-to-object snapping
 - Equal spacing guides
 - Smart distance indicator
 - Snap to print area
 - Snap toggle
-

16. Print Area

আমাদের existing `CustomizerConfig.views[]`.printArea use করবে।

Canvas display করবে:

- Print area boundary

- Safe area
- Bleed area
- Cut line
- Trim line
- Product boundary
- Hide/show guidelines

Legend:

Red = Bleed
 Black = Cut / Trim
 Green = Safe Area

17. Print Safety Validation

User object ভুল জায়গায় রাখলে warning:

- Outside print area
- Outside safe area
- Object crosses trim line
- Text too close to cut
- Low-resolution image
- Image too small
- Object clipped
- Empty required print side

Example:

⚠ Your logo is outside the safe print area.

18. Product Views / Sides

Existing `CustomizerConfig.views[]` structure use করবে।

Possible views:

Flat Print

- Front
- Back

Apparel

- Front
- Back
- Left Sleeve
- Right Sleeve
- Chest
- Pocket

Mugs / Bottles

- Full Wrap
- Front
- Back
- Left
- Right

Boxes

- Front
- Back
- Left
- Right
- Top
- Bottom
- Full Dieline

Custom Products

Admin can define unlimited views.

19. View Switching

- Front/back tabs
- View thumbnails
- Save current view on switch
- Load next view
- Object count per view

- Preview per view
 - Unsaved-state indicator
 - Add custom side
-

20. Canvas Size & Dimensions

For products such as:

Business Cards

Flyers

Banners

Labels

Posters

Packaging

controls:

- Width
- Height
- Unit
- Orientation
- Portrait
- Landscape
- Lock aspect ratio
- Preset sizes
- Custom size

Units:

px

mm

cm

inch

Canvas logical pixel size আর physical print size আলাদা থাকবে।

21. Rulers & Measurements

Professional print editor-এর জন্য:

- Top ruler
 - Left ruler
 - Inch ruler
 - Millimeter ruler
 - Guide creation
 - Drag custom guides
 - Clear guides
 - Object X position
 - Object Y position
 - Object width
 - Object height
-

22. Position & Transform Panel

Selected object-এর জন্য:

- X coordinate
 - Y coordinate
 - Width
 - Height
 - Rotation
 - Lock ratio
 - Opacity
-

23. Templates

Admin-created design templates।

- Template browser
- Template categories
- Search templates
- Apply template
- Replace current design

- Template preview
- Recently used
- Favorite template

Templates product-specific হবে।

Example:

Business Card Templates

T-Shirt Templates

Mug Templates

Flyer Templates

Label Templates

24. Product Variants

Canvas-এর সাথে variant UI integrated থাকবে।

Options:

- Color
- Size
- Material
- Finish
- Orientation
- Product style
- Quantity

Example:

T-Shirt

Color: Black

Size: XL

Variant change করলে design state হারাতে না।

25. Dynamic Product Color

Apparel/product mockup-এর ক্ষেত্রে:

- Product color selector
 - Mockup color update
 - Print visibility validation
 - Color contrast warning
-

26. Save Design

Existing **designs** + **design_views** database use করবে।

- Save manually
 - Save button state
 - Save progress
 - Last saved timestamp
 - Save name/title
 - Save as new
 - Duplicate design
-

27. Autosave

- Debounced autosave
- Local IndexedDB backup
- Server autosave
- Autosave indicator

Example:

Saving...

Saved

All changes saved

28. Design Recovery

- Browser crash recovery
 - Restore local draft
 - Restore last server version
 - Unsaved changes warning
-

29. Design Versioning

Existing:

design_versions
design_version_views

use করবে।

- Manual save version
 - Add-to-cart version
 - Ordered version
 - Restore version
 - View version history
-

30. Preview

Instant Preview

- Current canvas preview
- Front preview
- Back preview
- All-side preview

Mockup Preview

- Product mockup
 - Actual product color
 - Design placement
 - Fullscreen preview
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31. 3D Preview — Advanced Phase

Later:

- 3D T-shirt
- 3D Mug
- 3D Bottle
- 3D Box
- Rotate product
- Zoom product
- CanvasTexture mapping

Technology:

Three.js

React Three Fiber

Drei

32. Price Calculator

Backend pricing engine-এর result display করবে।

Canvas modifications অনুযায়ী:

- Base product price
- Extra print side
- Premium option
- Background removal
- Special finish
- Quantity
- Discount
- Final total

Example:

Base Product	\$20
Back Printing	\$5
Background Removal	\$3

Total	\$28

Frontend price calculate করবে না—backend source of truth।

33. Quantity

- Quantity selector
 - Quantity input
 - Min quantity
 - Max quantity
 - Bulk quantity
 - Quantity discount
-

34. Add to Cart Validation

Add to Cart করার আগে:

- Product selected
 - Variant selected
 - Required sides complete
 - No fatal print warning
 - Design saved
 - Design version locked
 - Price verified
 - Preview generated
-

35. File Download — Customer

If enabled:

- Download preview
- Download proof
- PDF proof
- Low-resolution watermarked proof

Production print file customer-কে defaultভাবে দেওয়া হবে না।

36. Admin Print Configuration

Admin Customizer Config-এর মধ্যে:

- Physical width
- Physical height
- DPI
- Bleed
- Safe area
- Print method
- File format
- Color mode
- Transparent export

Print methods:

Digital Print

Offset Print

DTG

DTF

Screen Print

Sublimation

Embroidery

UV Print

Laser Engraving

Vinyl

37. Flat Print Mode

Products:

Business Cards

Flyers

Brochures

Postcards

Labels

Stickers

Banners

Posters
Tickets
Menus

Features:

- Fixed canvas size
 - Front/back
 - Bleed
 - Trim
 - Safe area
 - DPI validation
 - PDF export
-

38. Apparel Mode

Products:

T-Shirts
Hoodies
Polo Shirts
Caps
Aprons
Tote Bags

Features:

- Multiple print locations
 - Product color
 - Front/back
 - Sleeve
 - Print-size restrictions
 - Product mockup
 - Embroidery area — optional
-

39. Wrap Mode

Products:

Mugs
Tumblers
Water Bottles
Travel Mugs
Sipper Bottles

Features:

- Wrap canvas
 - Seam area
 - Left/right boundaries
 - Full-wrap preview
 - Front/back preview
 - Curved product preview
-

40. Packaging / Dieline Mode

Products:

Boxes
Cartons
Product Sleeves
Paper Bags

Features:

- Dieline overlay
 - Fold lines
 - Cut lines
 - Glue areas
 - Panel labels
 - Locked structural layer
 - Front/back/side panels
 - Full dieline export
-

41. Multipage Mode

Products:

Booklets
Catalogs
Magazines
Annual Reports
Notepads

Additional features:

- Add page
 - Delete page
 - Duplicate page
 - Reorder pages
 - Page thumbnails
 - Master template
 - Cover
 - Inside pages
 - Back cover
 - Multi-page PDF export
-

42. Forms Mode

Products:

NCR Forms
Invoices
Bill Books

Features:

- Fixed fields
- Editable fields
- Sequential numbering
- Multiple copies
- Duplicate sheets
- Perforation guide

- Numbering preview
-

43. Photo Product Mode

Products:

Photo Frames

Cushions

Canvas Prints

Acrylic Blocks

Phone Cases

Features:

- Photo upload
 - Crop
 - Fit
 - Fill
 - Mask
 - Auto-center
 - Background options
 - Product preview
-

44. Keyboard Shortcuts

Professional UX:

Delete	→ Delete
Ctrl + Z	→ Undo
Ctrl + Shift + Z	→ Redo
Ctrl + C	→ Copy
Ctrl + V	→ Paste
Ctrl + D	→ Duplicate
Ctrl + A	→ Select All
Arrow Keys	→ Move
Shift + Arrow	→ Larger move

45. Mobile Touch Features

- Single finger move
- Pinch zoom
- Two-finger pan
- Touch resize handles
- Touch rotation
- Bottom toolbar
- Slide-up settings
- Sticky Add to Cart

46. Accessibility

- Keyboard navigation
- Proper labels
- ARIA attributes
- Focus states
- Sufficient contrast
- Tooltip descriptions

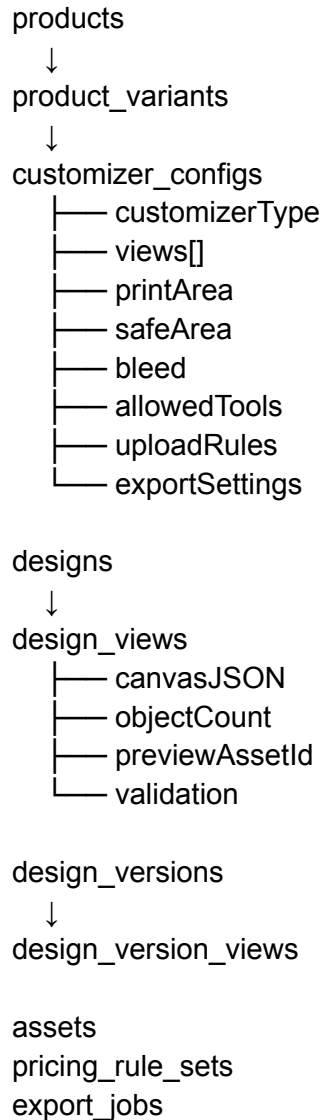
47. AI Features — Later Phase

Optional advanced:

- AI image generation
 - AI background removal
 - AI image enhancement
 - AI upscale
 - AI design suggestions
 - AI text generation
 - AI auto-layout
 - AI print-quality checker
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48. Current Database Mapping

Canvas features existing DB structure-এর সাথে এভাবে map হবে:



49. Recommended

CustomizerConfig.allowedTools

Database-এ এই ধরনের config রাখলে product অনুযায়ী tools enable/disable করা যাবে:

text
uploadImage
shapes
clipart
qrCode
barcode
background
patterns
layers
crop
imageFilters
backgroundRemoval
curvedText
templates
multipage
dieline

Example:

Business Card

Text 
Image 
Shapes 
QR 
Barcode 
Layers 
Multipage 
Dieline 

আর:

Corrugated Box

Text 
Image 
Shapes 
Layers 
Dieline 
Multipage 

50. Recommended Editor Types

আমাদের existing architecture-এ এই enum add করো:

flat
apparel
wrap
packaging
multipage
form
photo

Mapping:

Business Card → flat
T-Shirt → apparel
Mug → wrap
Box → packaging
Magazine → multipage
NCR Form → form
Phone Case → photo

Recommended Development Priority

Phase 1 — Core Canvas

- Canvas
- Text
- Image
- Shapes
- Drag
- Resize
- Rotate
- Delete
- Layers
- Undo/Redo
- Print area
- Front/back

- Save/restore

Phase 2 — Professional Editor

- Smart alignment
- QR
- Clipart
- Templates
- Crop
- Image filters
- Rulers
- Guides
- Design validation
- Autosave

Phase 3 — Product-Specific Editors

- Apparel
- Wrap
- Packaging
- Multipage
- Form
- Photo

Phase 4 — Advanced

- Background removal
 - AI
 - 3D
 - Advanced PDF/export
 - Collaborative editing
-

Final Canvas UI Structure

এই list অনুযায়ী build করলে আমাদের customizer শুধু T-shirt বা business card editor হবে না; একই core engine দিয়ে **printing, apparel, drinkware, packaging, signage, photo products, multipage documents** এবং **corporate gifts** handle করা যাবে।